

# Mobile and Embedded Computing

**Lecture 5.** GraphQL & REST, Observability for Mobile applications (Events, Crashlytics, Performance Monitoring)

Section 1

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# Mobile API Communication

# Why APIs are Critical

-  **Data Source:** Apps are often containers; the data lives on remote servers.
-  **Functionality:** Enable complex features like login, payments, and social feeds.
-  **Synchronization:** Keep user data consistent across multiple devices (phone, tablet, web).
-  **Separation of Concerns:** Decouples the client (app) from the backend (server).

# REST API Fundamentals

## Architecture

**RE**presentational **S**tate **T**ransfer. A stateless, client-server architectural style.

It is not a protocol, but a set of constraints.

## Core Concepts

Uses standard HTTP methods to act on "Resources" identified by URLs.

GET /users/123

POST /users

PUT /users/123

DELETE /users/123

# REST in Mobile: Pros & Cons



## Pros

**Simplicity:** Easy to understand, built on standard HTTP.

**Caching:** Standard HTTP caching is effective.

**Mature:** Decades of tooling and support.



## Cons

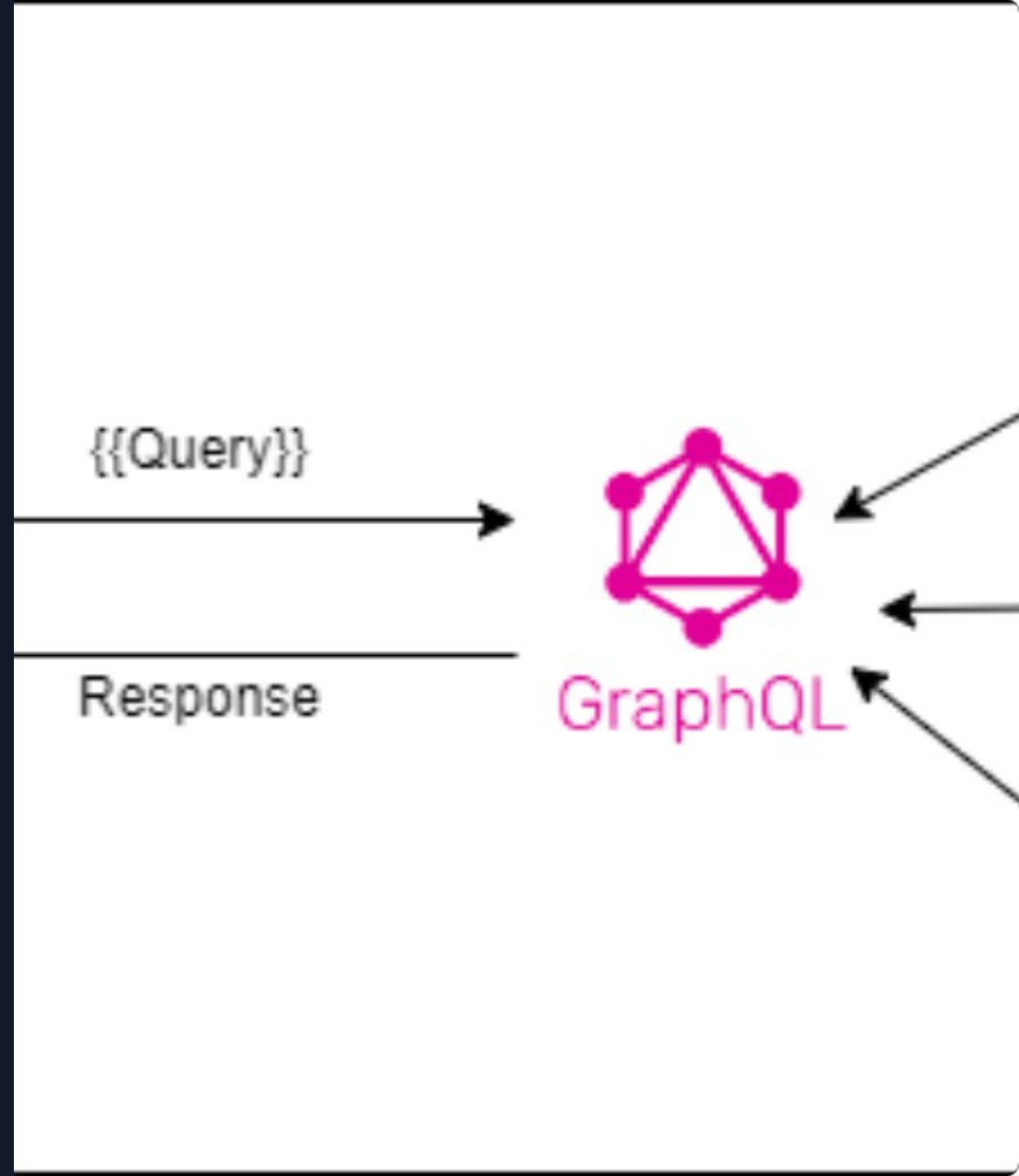
**Over-fetching:** Getting more data than the screen needs.

**Under-fetching:** Needing multiple API calls for one screen (e.g., user + posts).

# Introduction to GraphQL

A **query language** for your API, not an architecture. Developed by Facebook to solve mobile data-fetching issues.

Allows the client to ask for *exactly* what it needs and get *exactly* that in a single request.



# GraphQL Core Concepts



## Schema

The "contract." A strongly-typed definition of all data available on the API (types, fields, relationships).



## Query

A read-only operation to fetch data.  
The query's shape mirrors the JSON response's shape.



## Mutation

A write operation to modify data (create, update, delete). Defined in the schema just like queries.

# GraphQL vs. REST: Head-to-Head

| Feature       | REST                                     | GraphQL                                     |
|---------------|------------------------------------------|---------------------------------------------|
| Endpoint      | Multiple endpoints (/users, /posts)      | Single endpoint (e.g., /graphql)            |
| Data Fetching | Fixed data structure (defined by server) | Flexible data shape (defined by client)     |
| Data Shape    | Prone to over/under-fetching             | Precise data in one request                 |
| Schema        | Loosely defined (e.g., OpenAPI)          | Strongly typed (Schema Definition Language) |
| Protocol      | Uses all HTTP verbs (GET, POST, PUT...)  | Typically uses POST for all operations      |



# When to Use...

## Use REST when...

- ...simplicity is key.
- ...you have simple, well-defined resources.
- ...caching is a top priority (HTTP caching works perfectly).
- ...you are integrating with many 3rd-party APIs.

## Use GraphQL when...

- ...you have complex data relationships (e.g., social feeds).
- ...mobile bandwidth is a major concern (reduces payload).
- ...your app needs to fetch data from multiple sources.
- ...you have multiple clients (iOS, Android, Web) with different data needs.

Section 2

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# Mobile App Observability

# What is Mobile Observability?



**More than monitoring:** Monitoring tells you *\*if\** something is broken. Observability helps you understand *\*why\**.



**Three Pillars:** Traditionally Logs, Metrics, and Traces.



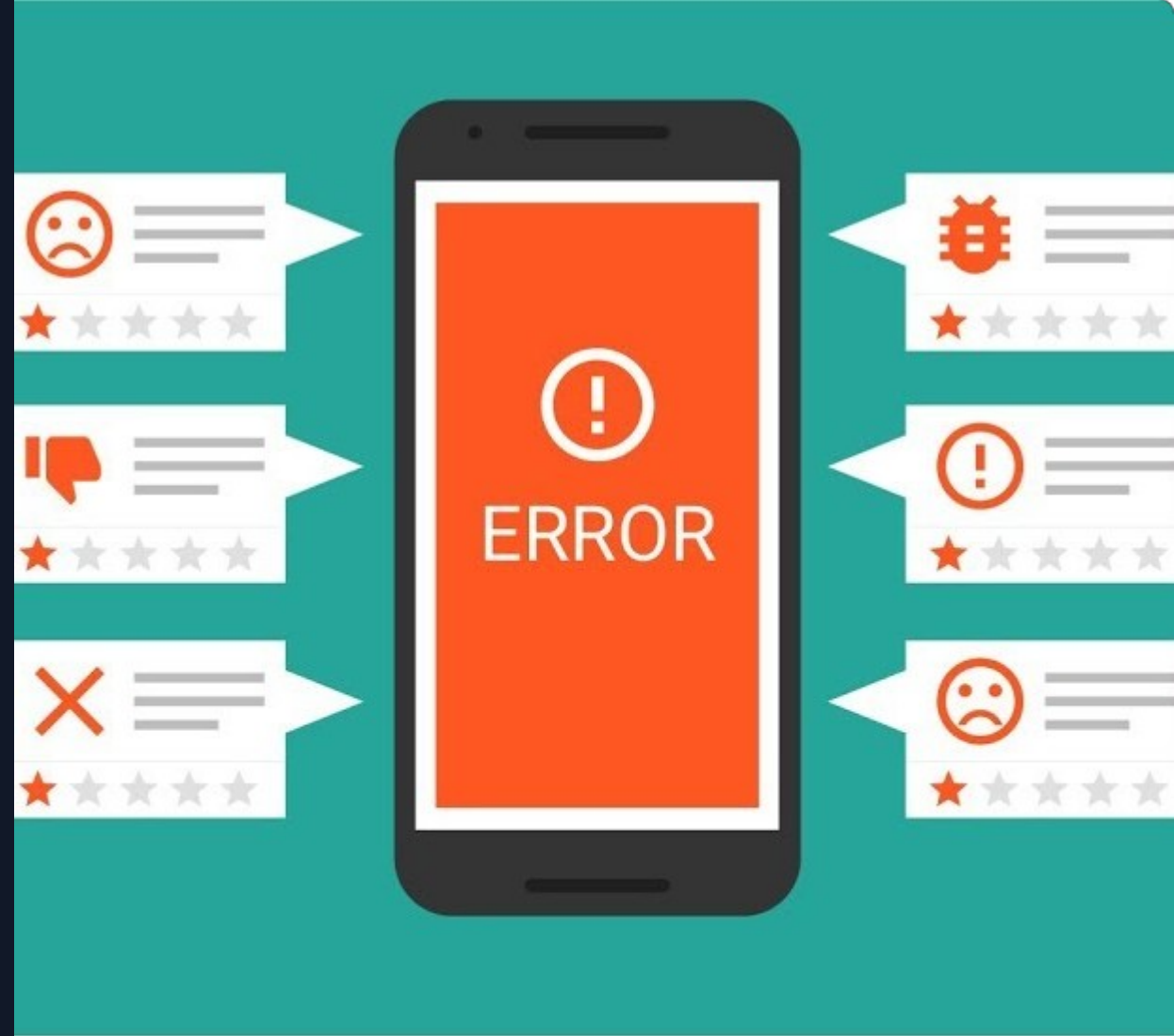
**For Mobile:** We adapt this to our "Big 3":

1. **Events** (User Behavior)
2. **Crashes** (Stability)
3. **Performance** (Speed)

# The Firebase Toolkit

Google's mobile platform that provides a suite of tools for building, releasing, and monitoring apps.

We focus on its observability products, which work together to give a complete picture of app health.



# Google Firebase Crashlytics

# 1. Firebase Analytics (Events)



**What it is:** Tracks user behavior and app usage by logging "Events".



**Automatic Events:** Captures screen\_view, app\_remove, first\_open automatically.



**Custom Events:** You log events that matter to your app (e.g., purchase\_complete, share\_image).



**The "Why":** Understand user funnels, feature adoption, and engagement.

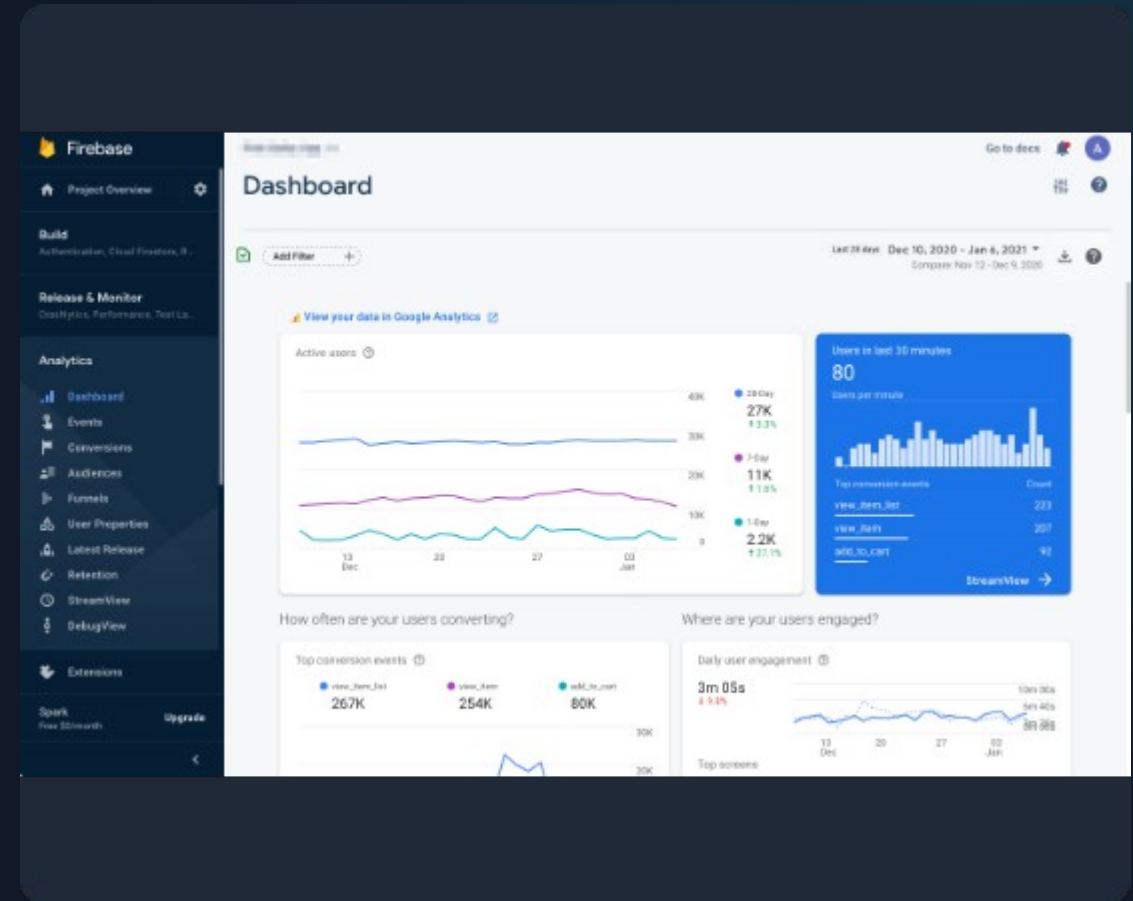
# Implementing Firebase Events

1. Get the FirebaseAnalytics instance:

```
private FirebaseAnalytics analytics;  
analytics = FirebaseAnalytics.getInstance(this);
```

2. Log your custom event with a Bundle:

```
Bundle bundle = new Bundle();  
bundle.putString("item_name", "Red T-Shirt");  
bundle.putString("category", "Apparel");  
analytics.logEvent("select_content", bundle);
```



## 2. Firebase Crashlytics

**99.9%**  
Goal: Crash-Free Users

### Find, Fix, and Prioritize Crashes

Real-time crash reporting that automatically captures and groups crashes (Java/Kotlin & NDK).

Provides actionable insights: stack traces, device info, and OS version.

# Setting Up Crashlytics



**Integration:** Add the Gradle dependency. It works out-of-the-box.



**Custom Logs:** Add "breadcrumb" context to reports.  
`Crashlytics.log("User at checkout screen");`



**Custom Keys:** Set user state (e.g., premium/free).  
`Crashlytics.setCustomKey("user_tier", "premium");`



**Test Crash:** Verify your setup.  
`throw new RuntimeException("Test Crash");`



### 3. Firebase Performance



#### App Startup

Measures "cold," "warm," and "hot" app start times. A critical first impression.



#### Network Requests

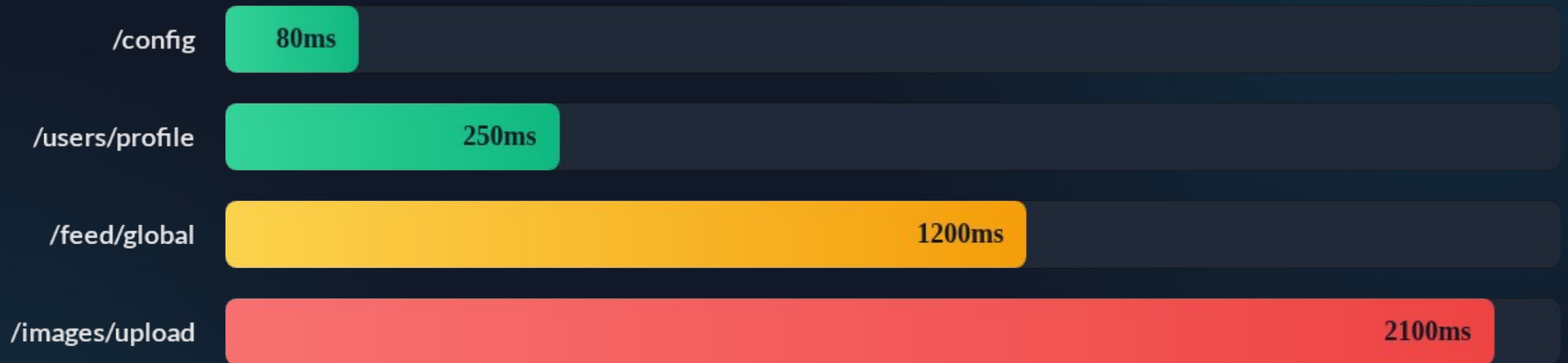
Automatically monitors HTTP/S requests. Tracks success rate, payload size, and response time.



#### Custom Traces

Manually trace specific code blocks (e.g., "image\_processing") to find bottlenecks.

# Performance Deep Dive: Network



*The /feed/global and /images/upload endpoints are clear bottlenecks that need investigation.*

# Real life example



## 1. Event

User clicks "Buy". (Analytics logs  
begin\_checkout).



## 2. Performance

checkout\_trace starts. Network call to  
/payment is **slow (5.2s)**. (Performance  
alerts).



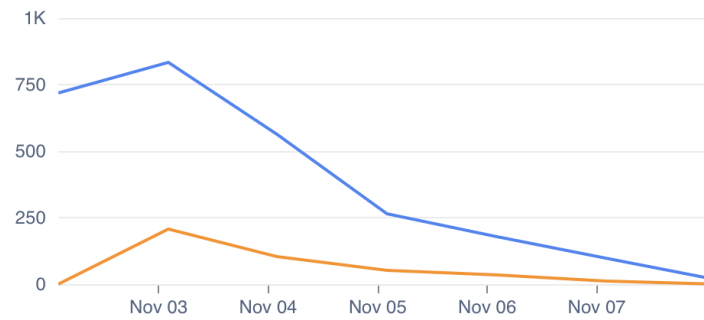
## 3. Crash

App crashes due to an unexpected  
null response from the slow /payment  
API. (Crashlytics reports).

## Total events by version

for the last 7 days

- 4.7.11 (436)  
**2.7K**
- 4.7.15 (440)  
**415**



## Devices

63% Samsung

## Operating systems

37% Android 15

## Device states

2% background

63% Samsung →

13% Xiaomi →

8% Motorola →

5% Oppo →

11% Other (5)

## Events



Alert your team when an important issue like this one occurs by setting up an integration

See integrations



## Event summary

4.7.11 (436)

16

Galaxy S24 Ultra

Nov 9, 2025, 10:32:39 AM

## Stack trace

Keys

Logs & Breadcrumbs

Data

Rollouts



TXT



Fatal Exception: io.flutter.plugins.firebase.crashlytics.FlutterError

Provider<ConsentsCubit> not found for ScannerAlimente



```
Fatal Exception: io.flutter.plugins.firebase.crashlytics.FlutterError: Provider<ConsentsCubit> not found for ScannerAlimente
at Provider._inheritedElementOf(provider.dart:377)
at Provider.of(provider.dart:327)
at ReadContext.read(provider.dart:683)
at _ScannerAlimenteState._checkAndPromptConsents(main.dart:657)
at _ScannerAlimenteState._initApp.<fn>(main.dart:526)
at SchedulerBinding._invokeFrameCallback(binding.dart:1434)
at SchedulerBinding.handleDrawFrame(binding.dart:1361)
at SchedulerBinding._handleDrawFrame(binding.dart:1200)
```

Event summary

4.7.11 (436)

16

Galaxy S24 Ultra

Nov 9, 2025, 10:32:39 AM

<

>

Stack trace

Keys

Logs & Breadcrumbs

Data

Rollouts

All

Logs

Events

Filter logs

Expand all

Download logs

| #   | Time (newest first) ↑ | Source                                                                            | Log           |
|-----|-----------------------|-----------------------------------------------------------------------------------|---------------|
| ▶ 1 | 10:32:38.383 AM       |  | app_open      |
| ▶ 2 | 10:32:38.160 AM       |  | screen_view   |
| ▶ 3 | 10:32:38.080 AM       |  | session_start |

Items per page: 25 1 - 3 of 3 < >

Percentile  
90%

Filter

Last 7 days  
Nov 3 – Nov 9

App start time

App start time (1.30s) is **8% faster** compared to 6 days prior



● All versions

● 4.7.31 (456)

Select version

1.30s

1.09s

—

View metric details

App start time

-8% 1.30s

Duration

find-codes-in-background

+84% 188 ms

Duration

vertex-ai-generation

-25% 6.96s

Success rate

scan.cloud.codingshadows.com/\*\*

+8 97.22%

Response time

scan.cloud.codingshadows.com/\*\*

-26% 1.76s

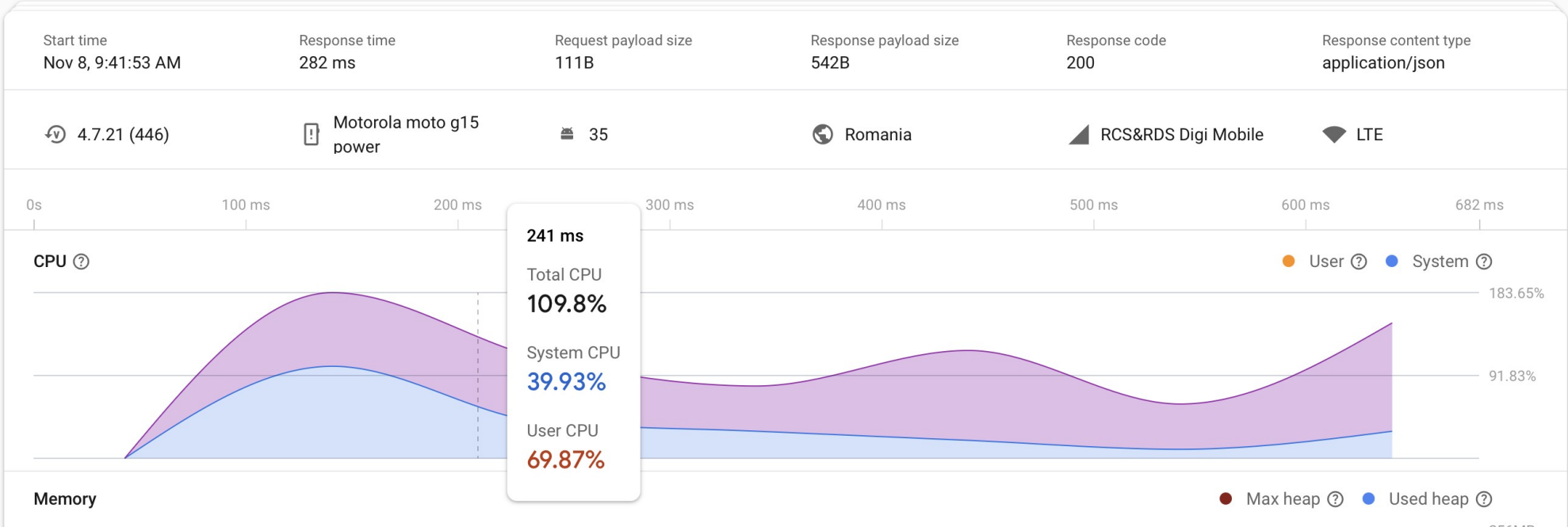
Select a metric

+

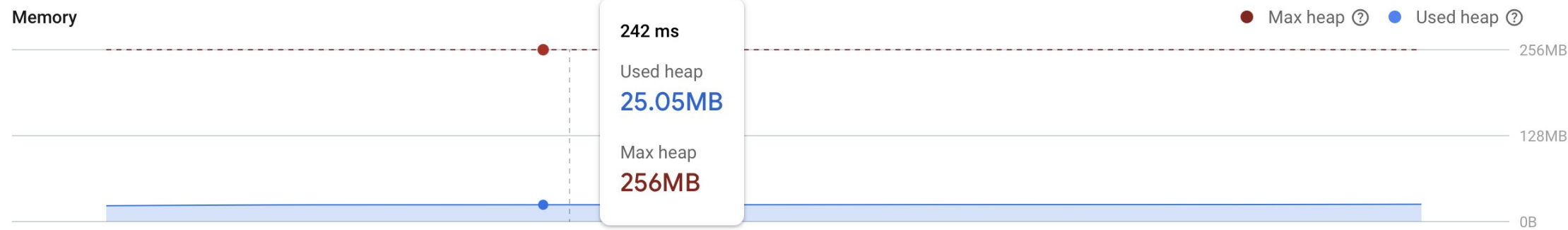
Distribution ?



Sessions (9 of 811)



## Memory



## On device

product\_prices\_fetch

## Network

scan.cloud.codingshadows.com...

scan.cloud.codingshadows.com/\*\*



◆ Gemini

| Requests | Success rate | Latency (p95) | Tokens |
|----------|--------------|---------------|--------|
| 3.7K     | 100%         | 3.61s         | 6.8M   |

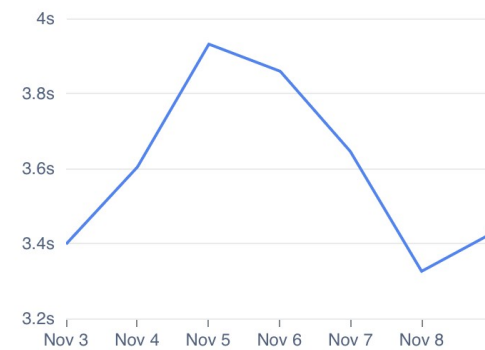
Requests



Success rate



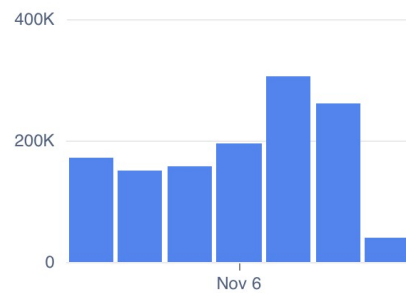
Latency (p95)



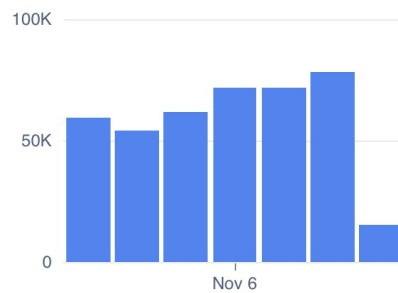
Tokens

Text Image Video Audio Document

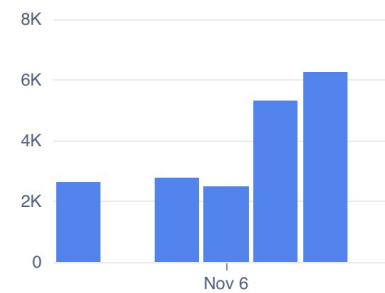
Input  
1.3M



Output  
412K



Thinking  
18K



## Traces

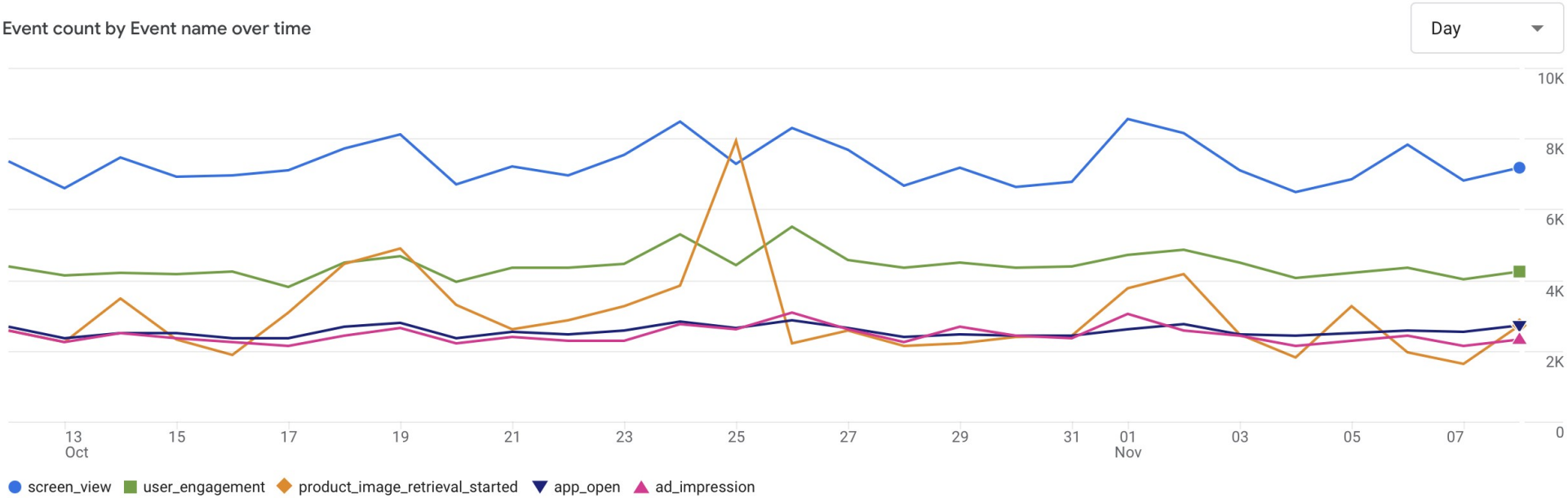
Sampling rate for all apps is 100%. [Edit rate](#) (?)

 **Filter** Add filter...

| Status | Created                  | Input                                                                                                                                        | Output                                                                                                                             | App version | Latency | Model                 |
|--------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-------------|---------|-----------------------|
| ✔      | Nov 9, 2025, 10:50:07 AM | {\"content\":<br>{\"data\":\"image/jpeg<sha5121165cfe1ae3aa32f434063afe729054498d6700ff938d78c4c53dd92f2591431d097c56934e277c5f9fd9795ed...  | {\"event.name\":\"gen_ai.choice\",\"finish_reason\":\"stop\",\"index\":0,\"message\":{\"content\":{\"text\":\"\\n...               | unknown     | 1.56s   | gemini-2.5-flash-lite |
| ✔      | Nov 9, 2025, 10:47:15 AM | {\"content\":<br>{\"data\":\"image/jpeg<sha5128ee129e69ba6c8129e1e6b387e9357c4b7728e0c75f54f54da05f5751d914b3f0ea4d963a7365d9fa0f11ddb81...  | {\"event.name\":\"gen_ai.choice\",\"finish_reason\":\"stop\",\"index\":0,\"message\":{\"content\":{\"text\":\"\\n...               | unknown     | 1.61s   | gemini-2.5-flash-lite |
| ✔      | Nov 9, 2025, 10:47:02 AM | {\"content\":<br>{\"data\":\"image/jpeg<sha5120cb4ec7c4558b6a094afc145e85ddb7ea77d6269581aab2859b001f2bcd11755f9f7f8c33041ab8a9d88a44565...  | {\"event.name\":\"gen_ai.choice\",\"finish_reason\":\"stop\",\"index\":0,\"message\":{\"content\":{\"text\":\"\\n...               | unknown     | 2.01s   | gemini-2.5-flash-lite |
| ✔      | Nov 9, 2025, 10:44:30 AM | {\"content\":<br>{\"data\":\"image/jpeg<sha512cf3096b4cfb096bdbfa2dbe1ba637b02202f5925277be45842e082b721ea0b72e225f5505375917023e567f96...   | {\"event.name\":\"gen_ai.choice\",\"finish_reason\":\"stop\",\"index\":0,\"message\":{\"content\":{\"text\":\"\\n...               | unknown     | 1.53s   | gemini-2.5-flash-lite |
| ✔      | Nov 9, 2025, 10:42:40 AM | {\"content\":<br>{\"data\":\"image/jpeg<sha5123fbeda4f9ecbe80d737eae6203a1f96c83e4bbcedae4c30b516a08cf272d956990e55da0f94e1af7d0c30ba5439... | {\"event.name\":\"gen_ai.choice\",\"finish_reason\":\"stop\",\"index\":0,\"message\":{\"content\":{\"text\":\"lapte pasteurizat... | unknown     | 1.46s   | gemini-2.5-flash-lite |

Items per page: 5 ▼ ‹ ›

Event count by Event name over time



Plot rows

Search...

Rows per page: 10 ▾ Go to: 1 < 1-10 of 258 >

|                                     | Event name                                        | Event count                | Total users             | Event count per active user | Total revenue             |   |
|-------------------------------------|---------------------------------------------------|----------------------------|-------------------------|-----------------------------|---------------------------|---|
| <input type="checkbox"/>            | Total                                             | 1,471,901<br>100% of total | 25,444<br>100% of total | 72.16<br>Avg 0%             | \$923.94<br>100% of total |   |
| <input checked="" type="checkbox"/> | 1 <a href="#">screen_view</a>                     | 204,544 (13.9%)            | 21,018 (82.6%)          | 10.13                       | \$0.00 (0%)               | ⋮ |
| <input checked="" type="checkbox"/> | 2 <a href="#">user_engagement</a>                 | 123,492 (8.39%)            | 19,990 (78.56%)         | 6.18                        | \$0.00 (0%)               | ⋮ |
| <input checked="" type="checkbox"/> | 3 <a href="#">product_image_retrieval_started</a> | 84,300 (5.73%)             | 2,310 (9.08%)           | 36.83                       | \$0.00 (0%)               | ⋮ |
| <input checked="" type="checkbox"/> | 4 <a href="#">app_open</a>                        | 71,411 (4.85%)             | 20,944 (82.31%)         | 3.54                        | \$0.00 (0%)               | ⋮ |